

SIMATS ENGINEERING

CO3-ASSESSMENT TOOL 1

Experiments

COURSE NAME: Computer Networks

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO3: *Analyze the performance of core network protocols such as TCP, UDP, HTTP, and DNS under varying conditions*

Assessment Tool Weightage: 40%

Dr.V.Parthipan

Code of Conduct:

I GODDAVULA NITISH KANTH (192512232) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student : NITISH KANTH.G

Criteria	<u>Understanding of Concepts</u> 2.5	Experimental Design 3	Use of Tools 2	Analysis of Results 1.5	Documentation 1	<u>Total</u> (10)
<u>Q1</u>	2	2	2	1.5	1	8.5
<u>Q2</u>	2	2	2	1.5	1	8.5
<u>Q3</u>	2	2	2	1	1	8
<u>Q4</u>	2	2	2	1.5	1	8.5

Faculty Signature

Experiments

QUESTIONS:4

Total Marks:40

1. Capture packets during a TCP connection setup. Identify the three packets involved in the handshake (SYN, SYN-ACK, and ACK). Demonstrate the role of each flag and how they establish a reliable connection. (BL4) (10)
2. Capture a DNS query using UDP in Wireshark. Compare the UDP header size with TCP and note the absence of sequence numbers or acknowledgments. Discuss how this lightweight design impacts speed and reliability. (BL4) (10)
3. Simulate packet loss (e.g., disconnect briefly) and capture traffic in Wireshark. Observe how TCP retransmits lost packets while UDP does not. Explain why this difference makes TCP reliable but slower compared to UDP. (BL4) (10)
4. Use Wireshark to capture DNS traffic while resolving a domain name. Identify the query type (A, AAAA, MX) and the response received. Measure the time taken for resolution and explain why DNS typically uses UDP. (BL4) (10)

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

SOLUTIONS:

1)

Wireshark - Packet 1276 - Wi-Fi

Frame 1276: Packet, 55 bytes on wire (440 bits), 55 bytes captured (440 bits) on interface \Device\NPF_{93CA38C7-E8C4-4430-9C4A-1457C602}

Ethernet II, Src: Intel_97:ec:21 (4c:0f:3e:97:ec:21), Dst: Cisco_6d:bc:f7 (24:d5:e4:6d:bc:f7)

Internet Protocol Version 4, Src: 172.23.23.169, Dst: 192.178.211.188

Transmission Control Protocol, Src Port: 60690, Dst Port: 5228, Seq: 1, Ack: 1, Len: 1

Data (1 byte)

Source Port: 60690
Destination Port: 5228
[Stream index: 4]
[Stream Packet Number: 1]
[Conversation completeness: Incomplete (60)]
[TCP Segment Len: 1]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 2686945166
[Next Sequence Number: 2 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 1067029538
0101 = Header Length: 20 bytes (5)

0000 24 d5 e4 6d bc f7 4c 0f 3e 97 ec 21 08 00 45 00 \$ - . L > . ! . E
0010 00 29 3f f6 40 00 80 06 00 00 ac 17 17 a9 c0 b2) 9 @
0020 d3 bc ed 12 14 6c a0 27 87 8e 3f 99 94 22 50 10 ... 1 " P
0030 00 fd 58 4b 00 00 00 .. X

2)

Wireshark - Packet 1276 - Wi-Fi

Frame 1276: Packet, 55 bytes on wire (440 bits), 55 bytes captured (440 bits) on interface \Device\NPF_{93CA38C7-E8C4-4430-9C4A-1457C602}

Ethernet II, Src: Intel_97:ec:21 (4c:0f:3e:97:ec:21), Dst: Cisco_6d:bc:f7 (24:d5:e4:6d:bc:f7)

Internet Protocol Version 4, Src: 172.23.23.169, Dst: 192.178.211.188

Transmission Control Protocol, Src Port: 60690, Dst Port: 5228, Seq: 1, Ack: 1, Len: 1

Data (1 byte)

Source Port: 60690
Destination Port: 5228
[Stream index: 4]
[Stream Packet Number: 1]
[Conversation completeness: Incomplete (60)]
[TCP Segment Len: 1]
Sequence Number: 1 (relative sequence number)
Sequence Number (raw): 2686945166
[Next Sequence Number: 2 (relative sequence number)]
Acknowledgment Number: 1 (relative ack number)
Acknowledgment number (raw): 1067029538
0101 = Header Length: 20 bytes (5)

0000 24 d5 e4 6d bc f7 4c 0f 3e 97 ec 21 08 00 45 00 \$ - . L > . ! . E
0010 00 29 3f f6 40 00 80 06 00 00 ac 17 17 a9 c0 b2) 9 @
0020 d3 bc ed 12 14 6c a0 27 87 8e 3f 99 94 22 50 10 ... 1 " P
0030 00 fd 58 4b 00 00 00 .. X

3)

Wireshark - Packet 23 - Wi-Fi

No.	Time	Source	Destination	Protocol	Length	Info
7	0.011906600	43.175.213.92	172.23.23.169	TCP	58	443 → 54778 [ACK] Seq=1 Ack=1 Win=511 Len=0
9	0.012181400	43.175.213.92	172.23.23.169	TCP	58	443 → 54778 [ACK] Seq=1 Ack=766 Win=511 Len=0
15	0.017406600	43.175.213.92	172.23.23.169	TLSv1.2	659	Application Data
16	0.017408100	43.175.213.92	172.23.23.169	TLSv1.2	82	Application Data
17	0.017408900	43.175.213.92	172.23.23.169	TCP	58	443 → 54778 [FIN, ACK] Seq=626 Ack=766 Win=511 Len=0
18	0.017482700	172.23.23.169	43.175.213.92	TCP	54	54778 → 443 [ACK] Seq=766 Ack=627 Win=252 Len=0
19	0.018586400	172.23.23.169	43.175.213.92	TCP	54	54778 → 443 [FIN, ACK] Seq=766 Ack=627 Win=252 Len=0
20	0.018878200	43.175.213.92	172.23.23.169	TCP	58	443 → 59925 [ACK] Seq=1 Ack=1 Win=511 Len=0
21	0.018891700	43.175.213.92	172.23.23.169	TLSv1.2	353	Application Data
22	0.018893400	43.175.213.92	172.23.23.169	TLSv1.2	353	Application Data
23	0.018971300	172.23.23.169	43.175.213.92	TCP	54	59925 → 443 [ACK] Seq=763 Ack=591 Win=251 Len=0
24	0.019356000	172.23.23.169	43.175.213.92	TCP	66	57832 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
25	0.020636400	43.175.213.92	172.23.23.169	TCP	58	443 → 59925 [ACK] Seq=591 Ack=763 Win=511 Len=0
31	0.025071400	43.175.213.92	172.23.23.169	TLSv1.2	632	Application Data
32	0.025073200	43.175.213.92	172.23.23.169	TLSv1.2	82	Application Data
33	0.025074400	43.175.213.92	172.23.23.169	TCP	58	443 → 59925 [FIN, ACK] Seq=1189 Ack=763 Win=511 Len=0
34	0.025178000	172.23.23.169	43.175.213.92	TCP	54	59925 → 443 [ACK] Seq=763 Ack=1190 Win=251 Len=0
35	0.026428800	172.23.23.169	43.175.213.92	TCP	54	59925 → 443 [FIN, ACK] Seq=763 Ack=1190 Win=251 Len=0
36	0.028235500	172.23.23.169	43.175.213.92	TCP	66	53129 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
39	0.042094300	43.175.213.92	172.23.23.169	TCP	70	443 → 51265 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM

Frame 23: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF...

Ethernet II, Src: Intel 97:ec:21 (4c:0f:3e:97:ec:21), Dst: Cisco 6d:bc:f7 (24:d5:e4:6d:bc:f7)

Internet Protocol Version 4, Src: 172.23.23.169, Dst: 43.175.213.92

Transmission Control Protocol, Src Port: 59925, Dst Port: 443, Seq: 763, Ack: 591, Len: 0

Source Port: 59925
Destination Port: 443
[Stream index: 7]
[Stream Packet Number: 4]
[Conversation completeness: Incomplete (28)]
[TCP Segment Len: 0]
Sequence Number: 763 (relative sequence number)
Sequence Number (raw): 424216372
[Next Sequence Number: 763 (relative sequence number)]
Acknowledgment Number: 591 (relative ack number)
Acknowledgment number (raw): 3904980505
0101 = Header Length: 20 bytes (5)
Flags: 0x010 (ACK)
000. = Reserved: Not set
...0 = Accurate ECN: Not set
...0 = Congestion Window Reduced: Not set
...0 = ECN Echo: Not set
...0 = Urgent: Not set
...0 = Acknowledgment: Set
...0 = Push: Not set
...0 = Reset: Not set
...0 = Syn: Not set
...0 = Fin: Not set
[TCP Flags:A....]
Window: 253
0000 24 d5 e4 6d bc f7 4c 0f 3e 97 ec 21 08 00 45 00 \$ = L > I E
0010 00 2b db ee 40 00 00 06 00 00 ac 17 17 a9 2b af (.....+
0020 d5 5c ea 15 01 bb 19 49 07 34 e8 c1 46 19 50 10 \.....I .4..F.
0030 00 fd c4 e6 00 00

4)

Wireshark - Packet 1970 - Wi-Fi

No.	Time	Source	Destination	Protocol	Length	Info
1969	47.578311500	172.23.23.169	172.21.100.31	DNS	92	Standard query 0x8ce7 A mobile.events.data.microsoft.com
1970	47.598841200	172.21.100.31	172.23.23.169	DNS	212	Standard query response 0x8ce7 A mobile.events.data.microsoft.com CNAME mobile.events.data.trafficmanager.net CNAME onedscolprdeus19...
2737	67.679363200	172.23.23.169	172.21.100.31	DNS	84	Standard query 0x7e93 A dls.di.atlas.samsung.com
2738	67.699835200	172.21.100.31	172.23.23.169	DNS	122	Standard query response 0x7e93 A dls.di.atlas.samsung.com CNAME dls-gcp.di.atlas.samsung.com A 34.107.165.5
3084	75.712167200	172.23.23.169	172.21.100.31	DNS	70	Standard query 0xad8 HTTP/2 dns.google
3085	75.713997500	172.23.23.169	172.21.100.31	DNS	70	Standard query 0xa8a dns.google
3086	75.726244100	172.21.100.31	172.23.23.169	DNS	102	Standard query response 0xa8a A dns.google A 8.8.4.4 A 8.8.8.8
3087	75.726252100	172.21.100.31	172.23.23.169	DNS	102	Standard query response 0xa8a A dns.google A 8.8.4.4 A 8.8.8.8
3088	75.726292100	172.23.23.169	172.21.100.31	DNS	102	Standard query response 0xa8a A dns.google A 8.8.4.4 A 8.8.8.8
3089	75.731582500	172.21.100.31	172.23.23.169	DNS	102	Standard query response 0xa8a A dns.google A 8.8.4.4 A 8.8.8.8
3293	78.958048600	192.168.0.239	255.255.255.255	DNS	102	Standard query response 0xa8a A dns.google A 8.8.4.4 A 8.8.8.8

Frame 1970: Packet, 212 bytes on wire (1696 bits), 212 bytes captured (1696 bits) on interface \Device\NPF_{93CA30C7-E0C4-443D-9C4A-1457C60276E2}, id 0

Ethernet II, Src: Cisco 6d:bc:f7 (24:d5:e4:6d:bc:f7), Dst: Intel 97:ec:21 (4c:0f:3e:97:ec:21)

Internet Protocol Version 4, Src: 172.21.100.31, Dst: 172.23.23.169

User Datagram Protocol, Src Port: 53, Dst Port: 50963

Domain Name System (response)

Transaction ID: 0x8ce7

Flags: 0x8180 Standard query response, No error

Questions: 1
Answer RRs: 3
Authority RRs: 0
Additional RRs: 0

Queries

mobile.events.data.microsoft.com: type A, class IN

Answers

mobile.events.data.microsoft.com: type A, class IN

[Request Tx: 1969]
[Time: 20.529700 milliseconds]

0000 4c 0f 3e 97 ec 21 24 d5 e4 6d bc f7 08 00 45 00 L > I \$ m = E
0010 00 c6 16 34 00 00 76 11 51 de ac 15 64 if ac 17 ..T - Q . d .
0020 17 a9 00 35 c7 13 00 b2 fd 0e 8c e7 81 80 00 015.....
0030 00 03 00 00 00 00 06 6d 6f 62 69 6c 65 06 65 76m obile.ev
0040 65 6e 74 73 04 64 61 74 61 09 6d 69 63 72 6f 73 ents-dat a-micros
0050 6f 66 74 03 63 6f 6d 00 00 01 00 01 c0 0c 00 05 oft-com:
0060 00 01 00 00 00 56 00 27 06 6d 6f 62 69 6c 65 06V- .mobile
0070 65 76 65 6e 74 73 04 64 61 74 61 0e 74 72 61 66 events-d ata-traf
0080 66 69 63 6d 61 6e 61 67 65 72 03 6e 65 74 00 c0 ficmanag er-net..
0090 3e 00 05 00 01 00 00 00 28 00 29 10 6f 6e 65 64 >.....(.)oned
00a0 73 63 6f 6c 70 72 64 65 75 73 31 39 06 65 61 73 scolprde us19-eas
00b0 74 75 73 08 63 6e 6f 75 64 61 70 70 05 61 7e 75 tus-clou dapp-azu
00c0 72 65 c0 29 c0 71 00 01 00 01 00 00 00 08 00 04 re:).q:
00d0 34 a8 75 af 4-u.